Angled launches

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use components to analyze an angled projectile.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 8.*
10 m/s² downward. Timing resolution, sensor fit or release timing can contribute uncertainty.

02. Worked example: explain the model
Horizontal component 4 m/s throughout; vertical +10, 0,-10 m/s at the three marked times.

03. Read and interpret the model
Vertical velocity zero; horizontal remains; acceleration 10 m/s² down.

04. Apply the physics
1 s; 2 s; 12 m.

05. Explain your reasoning
Vertical motion is symmetric about top at equal height. A different landing height changes remaining fall time.

06. Review: connect an earlier idea *Review: Unit 1, Day 20.*
a=9 m/s² right; v_f=54 m/s right. Constant net force and mass throughout.

07. Exit check
1.5 s and 3 s.

08. STRETCH: test the model
Time and height unchanged; range doubles to 16 m under the same ideal assumptions.

Angled launches

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use components to analyze an angled projectile.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 8.*
$g=\Delta v/\Delta t=2.94/0.30=9.8$ m/s². Several points reveal a trend and reduce sensitivity to one noisy observation.

02. Worked example: explain the model
Horizontal component 4 m/s throughout; vertical +10, 0,-10 m/s at the three marked times.

03. Read and interpret the model
Vertical velocity zero; horizontal remains; acceleration 10 m/s² down.

04. Apply the physics
v_x 17.32, v_y 10 m/s; time 2 s; range 34.6 m; max height 5 m.

05. Explain your reasoning
Vertical motion is symmetric about top at equal height. A different landing height changes remaining fall time.

06. Review: connect an earlier idea *Review: Unit 1, Day 20.*
x interval [-77, -63] km; y interval [81, 99] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

07. Exit check
Both 34.6 m; 60° flight time 3.46 s versus 2 s at 30°. Complementary angles same ideal range.

08. STRETCH: test the model
$\sin(2\theta)=0.5$ gives 15° and 75°. Times $4\sin\theta\approx1.04$ and 3.86 s; same range, different arcs.